Spencer Killen

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Summary

My graduate studies have solidified my strong passion for symbolic artificial intelligence. Working as a teaching assistant has enabled me to discover my love for teaching.

Work Experience

Teaching Assistant (Part time) 2017-present

At the University of Alberta, I have been a teaching assistant for a cryptography course, a software engineering course, a course on search algorithms, and a course on non-procedural programming languages. My duties included marking, developing course material, presenting learning material, and helping students learn material.

Subreviewer 2019-present

Appointed by PC chairs, I have reviewed numerous submissions to conferences or journals related to my field of research as a subreviewer.

Education

Ongoing	PhD (Computing Science)	**University of Alberta**
2021	MSc (Computing Science)	**University of Alberta**
2019	BSc Honors (Computing Science) with distinction	**University of Alberta**
	BSc (Computer Science) (transferred out)	**MacEwan University**
2014	High School Diploma	**Wetaskiwin Composite High School**

Work Experience

Freelance Software Engineer (2015-2021)

As a freelance software engineer, I have worked with numerous companies to figure out their application needs and develop user-focused applications.

Scholarships

Alberta Innovates (2022-2025)

Alberta Innovates Graduate Student Scholarship

NSERC (2019-2020)

Natural Sciences and Engineering Research Council of Canada Canada Graduate Scholarships – Master's (NSERC CGS M), Scholarship

USRIPG (2017)

Undergraduate Student Research Initiative Project Grant, MacEwan University

Publications

Killen, Spencer and Jia-Huai You (2021a). "Fixpoint Characterizations of Disjunctive Hybrid MKNF Knowledge Bases". In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP)*.

Killen, Spencer and Jia-Huai You (2021b). “Unfounded Sets for Disjunctive Hybrid MKNF Knowledge Bases”. In: *Conference on Principles of Knowledge Representation and Reasoning, KR 2021.*

– (2022). “A Fixpoint Characterization of Three-Valued Disjunctive Hybrid MKNF Knowledge Bases”. In: *Conference on Logic Programming, ICLP 2022 Technical Communications.*

Killen, Spencer, Wenkai Gao, and Jia-Huai You (2023). “Expanding the Class of Polynomial Time Computable Well-Founded Semantics for Hybrid MKNF”. In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP).*

Killen, Spencer and Jia-Huai You (2024). “Adapting Approximation Fixpoint Theory to Nondeterministic Hybrid Reasoning”. In: *Workshop on Answer Set Programming and Other Computing Paradigms (ASPOCP).*

Kinahan, Riley et al. (2024). “On the Foundations of Conflict-Driven Solving for Hybrid MKNF Knowledge Bases”. In: *Theory Pract. Log. Program.*

Killen, Spencer and Jia-Huai You (2025a). “Eliminating Unintended Stable Fixpoints in Approximation Fixpoint Theory”. In: *Workshop on Non-Monotonic Reasoning (NMR 2025).*

– (2025b). “Using AFT to Characterize Shen and Eiter’s Disjunctive Logic Program Semantics”. In: *Workshop on Non-Monotonic Reasoning (NMR 2025).* CEUR Workshop Proceedings.

Killen, Spencer, Jia-Huai You, and Jesse Heyninck (2025). “An Alternative Theory of Stable Revision for Nondeterministic Approximation Fixpoint Theory and the Relationships”. In: *Association for the Advancement of Artificial Intelligence (AAAI).*